

Quiver-Unitary Invariance Characterizes Hyper-Reinhardt Free Spectrahedra

Automated mathematics run `fa_banach.001`, lane 10

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Status. Candidate full solution, likely valid, pending human review. The matrix-level invariance in Remark 4.4 of McCullough is proved equivalent to the defining path-block form of a hyper-Reinhardt pencil.

1 Source question

For a tuple $B = (B_1, \dots, B_g)$ on a finite-dimensional Hilbert space H , put

$$L_B(X) = I + \sum_{j=1}^g (B_j \otimes X_j + B_j^* \otimes X_j^*), \quad \mathfrak{P}_B[n] = \{X : L_B(X) \succ 0\}.$$

McCullough calls a free spectrahedron hyper-Reinhardt when it has a defining tuple of path-block form: there is an orthogonal decomposition $H = H_0 \oplus \dots \oplus H_g$ for which B_j has only its $(j-1, j)$ block nonzero. Lemma 4.3 of [1] shows that such a domain is invariant under

$$X_j \mapsto W_{j-1}^* X_j W_j, \quad W_0, \dots, W_g \in U(n). \tag{1}$$

Remark 4.4 asks whether this invariance characterizes the class.

The final $\mathfrak{P}_A$ in the source's displayed question is necessarily a local typo for $\mathfrak{P}_B$: A is not attached to the arbitrary domain being tested, and the preceding lemma makes the intended statement clear.

Remark 4.4. A natural question is whether a free spectrahedra $\mathfrak{P}_B \subseteq M(\mathbb{C})^g$ is hyper-Reinhardt if and only if for each n , each $X \in \mathfrak{P}_B[n]$ and tuple $W = (W_0, \dots, W_g) \in M_n(\mathbb{C})^{g+1}$ of unitary matrices

$$W \circ X = (W_0^* X_1 W_1, \dots, W_{g-1}^* X_g W_g) \in \mathfrak{P}_A.$$

Proof of Lemma 4.3. Letting D_W denote the block diagonal $(g+1) \times (g+1)$ with $(j+1, j+1)$ entry $I \otimes W_j$ for $0 \leq j \leq g$,

$$D_W^* L_A(T) D_W = L_A(W \circ T). \quad \blacksquare$$

Figure 1: Official arXiv:2107.11641v4 PDF page 17: all of Remark 4.4 and the block-conjugation proof of the forward implication.

2 Proof intuition

The symmetry is stronger than ordinary coordinatewise rotations because the W_j are full matrices and the right unitary on X_j is the left unitary on X_{j+1} . To exploit that strength, fix a matrix size n and replace the entries of each X_j by new free variables. The original and transformed pencils become two monic pencils in gn^2 variables. The hypothesis says their free spectrahedra agree at *all* auxiliary levels, so the Linear Gleichstellensatz makes their minimal coefficient tuples unitarily equivalent.

At $n = 2$, the identity $E_{11}E_{21} = 0$ can then be transported through this unitary equivalence. If two vertex unitaries in (1) are independent, choose their quotient to be the flip F ; now $E_{11}FE_{21} = E_{11} \neq 0$. The only way the transported product can still vanish is for its coefficient-space factor to vanish. Three placements of adjoints give exactly the orthogonality relations of a path representation. Their range and source spaces then reconstruct $H_0, \dots, H_g$.

3 Complete characterization

A monic defining tuple is *minimal* if no proper reducing subpencil defines the same free spectrahedron (equivalently, no smaller monic pencil does). Every free spectrahedron has a minimal defining tuple, unique up to unitary equivalence; we use the Linear Gleichstellensatz in this standard form [2, Proposition 2.2]. The same tool is used in the graph characterization of Reinhardt pencils in [3, Theorem 3.1].

Theorem 1 (answer to Remark 4.4). *Let $B = (B_1, \dots, B_g)$ be a minimal monic defining tuple, with every $B_j \neq 0$. The following are equivalent.*

1. *For every n , every $X \in \mathfrak{P}_B[n]$, and every $W_0, \dots, W_g \in U(n)$, the tuple in (1) belongs to $\mathfrak{P}_B[n]$.*
2. *There is an orthogonal decomposition*

$$H = H_0 \oplus H_1 \oplus \dots \oplus H_g$$

such that, with P_r the projection onto H_r ,

$$B_j = P_{j-1}B_jP_j \quad (1 \leq j \leq g). \quad (2)$$

Thus quiver-unitary invariance characterizes hyper-Reinhardt free spectrahedra. If the one-coordinate slices have radius one, the nonzero blocks in (2) have norm one, exactly matching the source's normalization.

We first verify the one technical point needed before applying the Gleichstellensatz.

Lemma 2 (matrix-unit inflation preserves minimality). *Let B be minimal and let $(E_{ab})_{a,b=1}^n$ be the matrix units of M_n . Then the tuple*

$$C_{j,ab} = B_j \otimes E_{ab} \quad (1 \leq j \leq g, 1 \leq a, b \leq n)$$

is a minimal defining tuple in gn^2 variables.

Proof. Decompose B into irreducible reducing summands $B^{(\nu)}$. Minimality means that no such summand is redundant. Fix one summand. By irreducibility, the unital C^* -algebra generated by its coefficients is the full matrix algebra M_{d_ν} . The span of positive-length $*$ -words in the $B_j^{(\nu)}$ is a nonzero two-sided ideal of M_{d_ν} , hence is all of M_{d_ν} .

Every positive-length $*$ -word Q can be lifted, using a compatible chain of matrix units, to $Q \otimes E_{ab}$ in the $*$ -algebra generated by the $C_{j,ab}^{(\nu)}$. Therefore that algebra contains $M_{d_\nu} \otimes M_n$ and the inflated summand is irreducible.

It remains to exclude redundancy among these summands. Restrict the new variables by setting $Y_{j,aa} = Z_j$ for every a and $Y_{j,ab} = 0$ for $a \neq b$. Then

$$\sum_{a,b} E_{ab} \otimes Y_{j,ab} = I_n \otimes Z_j,$$

and, after a tensor-factor permutation, each inflated summand becomes n copies of the original summand. Redundancy after inflation would therefore imply redundancy before inflation, a contradiction. Thus C is minimal. $\square$

Lemma 3 (forced products). *Under condition (1) of Theorem 1,*

$$B_j B_k = 0 \quad \text{if } k \neq j+1, \quad (3)$$

$$B_j B_k^* = 0 \quad \text{if } j \neq k, \quad (4)$$

$$B_j^* B_k = 0 \quad \text{if } j \neq k. \quad (5)$$

Proof. Fix n and write

$$X_j(Y) = \sum_{a,b=1}^n E_{ab} \otimes Y_{j,ab}.$$

The pencil $L_B(X(Y))$, after permuting tensor factors, has coefficients

$$C_{j,ab} = B_j \otimes E_{ab}.$$

For fixed $W_0, \dots, W_g \in U(n)$, the transformed pencil has coefficients

$$D_{j,ab} = B_j \otimes W_{j-1}^* E_{ab} W_j.$$

Condition (1), applied at sizes nm for every auxiliary size m , says precisely that C and D define the same free spectrahedron. Both are minimal by Lemma 2; hence the Linear Gleichstellensatz gives a unitary U_W such that

$$U_W C_{j,ab} U_W^* = D_{j,ab} \quad (6)$$

for every j, a, b .

Now take $n = 2$, let $F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and note

$$E_{11} E_{21} = 0, \quad E_{11} F E_{21} = E_{11} \neq 0. \quad (7)$$

If $k \neq j+1$, then W_j and W_{k-1} are independent choices. We have $C_{j,11} C_{k,21} = 0$; choose their quotient $W_j W_{k-1}^* = F$. By (7), the matrix factor of $D_{j,11} D_{k,21}$ is nonzero (the remaining outer factors are unitary). Equation (6) carries the zero product to a zero product, so $B_j B_k = 0$. This proves (3).

For $j \neq k$, start instead with

$$C_{j,11} C_{k,12}^* = 0 \quad \text{and choose } W_j W_k^* = F;$$

the same argument gives (4). Finally, start with

$$C_{j,11}^* C_{k,21} = 0 \quad \text{and choose } W_{j-1} W_{k-1}^* = F$$

to obtain (5). All choices are available exactly in the stated off-diagonal cases. $\square$

Proof of Theorem 1. Assume (1). Put

$$R_j = \text{ran } B_j, \quad S_j = \text{ran } B_j^*.$$

Relation (5) makes the R_j pairwise orthogonal, and (4) makes the S_j pairwise orthogonal. Moreover, (3) gives

$$R_k \perp S_j \quad \text{unless } k = j + 1,$$

because $\langle B_k x, B_j^* y \rangle = \langle B_j B_k x, y \rangle$. Consequently the subspaces

$$H_0 = R_1, \quad H_r = S_r + R_{r+1} \quad (1 \leq r \leq g-1), \quad H_g = S_g \quad (8)$$

are pairwise orthogonal.

Let K be the orthogonal complement of their sum. Since K is orthogonal to every S_j and every R_j , both B_j and B_j^* vanish on K . Thus K is a common reducing zero summand. If $K \neq 0$, deleting the identity block on K gives a smaller defining pencil, contrary to minimality. Hence the spaces in (8) sum to H .

In finite dimension, $(\ker B_j)^\perp = S_j$, so B_j vanishes on $S_j^\perp$ and has range R_j . Since $S_j \subseteq H_j$ and $R_j \subseteq H_{j-1}$, equation (2) follows. This proves (2).

Conversely assume (2) and set

$$\mathcal{W} = \bigoplus_{r=0}^g (I_{H_r} \otimes W_r).$$

A block computation gives

$$\mathcal{W}^* L_B(X) \mathcal{W} = L_B(W_0^* X_1 W_1, \dots, W_{g-1}^* X_g W_g).$$

Positive definiteness is therefore preserved, proving (1).

For the normalization statement, set all coordinates except X_j equal to zero. The only nontrivial block is

$$\begin{pmatrix} I & C_j \otimes X_j \\ C_j^* \otimes X_j^* & I \end{pmatrix}, \quad C_j = B_j|_{H_j} : H_j \rightarrow H_{j-1},$$

which is positive exactly when $\|C_j\| \|X_j\| < 1$. A coordinate-slice radius of one is therefore equivalent to $\|C_j\| = 1$. $\square$

Corollary 4 (inactive coordinates). *If some coordinate is absent from a minimal pencil, the characterization holds after deleting it, with zero connecting blocks reinserted. Thus the unqualified structural statement remains true if zero-dimensional layers or zero connectors are admitted; the source's norm-one convention simply excludes these unbounded coordinate directions.*

4 Verification, novelty, and limitations

The bundled exact checker verifies the 2×2 witness, the three families of zero-product relations on a nontrivial three-arrow path pencil, and the block-conjugation identity. These checks guard the index and adjoint placements; they do not replace Lemma 2 or the Gleichstellensatz.

Eight focused attempts are recorded in the run attempt note. A bounded literature search on 13 August 2026 covered the exact title, Remark 4.4 phrase, arXiv id, authors, and combinations of “hyper-Reinhardt”, “unitary invariance”, “quiver”, and “characterization”. No later claimed answer was found. McCullough–Tuovila [3] characterize only the weaker scalar torus symmetry. Ben Jemaa–McCullough [4] later use (1) as a necessary obstruction for a particular domain, but do not state the converse. Novelty confidence is moderate because the search was bounded.

Human review should focus on Lemma 2 and on the hypotheses of the Linear Gleichstellensatz. No conditional mathematical dependency is currently known. The only statement caveats are the source’s obvious A/B typo and its harmless norm-one coordinate normalization.

References

- [1] S. McCullough, *Hyper-Reinhardt Free Spectrahedra*, arXiv:2107.11641v4; published as a chapter in *Recent Developments in Operator Theory, Mathematical Physics and Complex Analysis*, 2023.
- [2] E. Evert, J. W. Helton, I. Klep, and S. McCullough, *Circular free spectrahedra*, J. Math. Anal. Appl. 445 (2017), 1047–1070; arXiv:1604.05756.
- [3] S. McCullough and N. Tuovila, *Reinhardt free spectrahedra*, Linear Algebra Appl. 640 (2022), 91–117; arXiv:2012.02289.
- [4] M. Ben Jemaa and S. McCullough, *Reinhardt Free Spectrahedra*, arXiv:2301.02746v2 (2023).